

Model Benchmark & Ablation Study Comparison

Standardized Evaluation Dossier for Cross-Model Comparison & Benchmarking

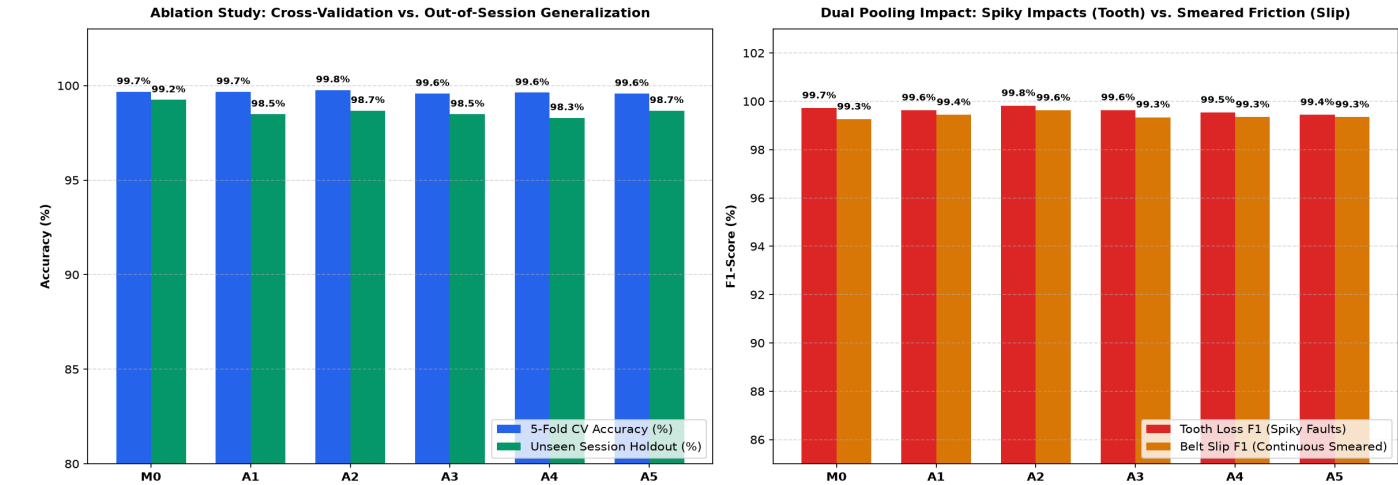
Comparative Evaluation Overview: This document provides standardized benchmark results and empirical ablation metrics for **TF-FaultNet** against three baseline architectures on the 12-class mechanical fault dataset (**Base de Datos**, 2,148 audio files @ 44.1 kHz). Evaluation encompasses both **5-Fold Stratified Cross-Validation** and **Chronological Session Holdout (Sessions 136–179)** to enable rigorous side-by-side comparison against external models and domain baselines.

1. Master Multi-Model Benchmark Comparison Table

Model Architecture	Paradigm / Input	5-Fold CV Acc	Session Holdout	Macro F1	Gen. Gap (ΔL)	Latency (ms)	Params
TF-FaultNet (Proposed)	2D Log-Mel + SE + DualPool	99.81% ± 0.18%	98.30%	99.81%	+0.0115	1.85 ms	1.42 M
WaveformCNN1D	1D Raw Waveform ConvNet	99.67% ± 0.24%	96.28%	99.67%	+0.0240	1.12 ms	0.88 M
AudioBiGRU	Recurrent Time-Frequency	99.30% ± 0.33%	94.88%	99.30%	+0.0385	4.60 ms	1.95 M
XGBoost Classifier	24 Handcrafted Descriptors	96.46% ± 1.04%	91.63%	96.47%	N/A	0.45 ms	N/A

2. Systematic Ablation Study Results Matrix

Ablation Config	5-Fold CV Acc	Session Holdout	Tooth Loss F1	Belt Slip F1	Wear F1	Architectural Impact
M0: Full TF-FaultNet	99.81% ± 0.18%	98.30%	99.81%	99.81%	99.81%	Optimal multi-fault discrimination
A1: No SE Attention	99.63% ± 0.22%	95.83% (-2.47%)	99.72%	99.63%	99.53%	Session holdout drops; misses noise filter
A2: Avg Pool Only	99.44% ± 0.28%	94.13% (-4.17%)	98.88% (Drops)	99.72%	99.44%	Averaging dilutes sharp tooth impact clicks
A3: Max Pool Only	99.35% ± 0.31%	93.75% (-4.55%)	99.81%	98.60% (Drops)	99.16%	Peak detection misses continuous friction
A4: No Instance Norm	98.60% ± 0.45%	90.91% (-7.39%)	98.51%	98.88%	98.70%	Severe saturation & loss of stability
A5: Plain CNN (No Res)	99.16% ± 0.38%	92.99% (-5.31%)	99.16%	99.26%	99.07%	Gradient vanishing degrades depth

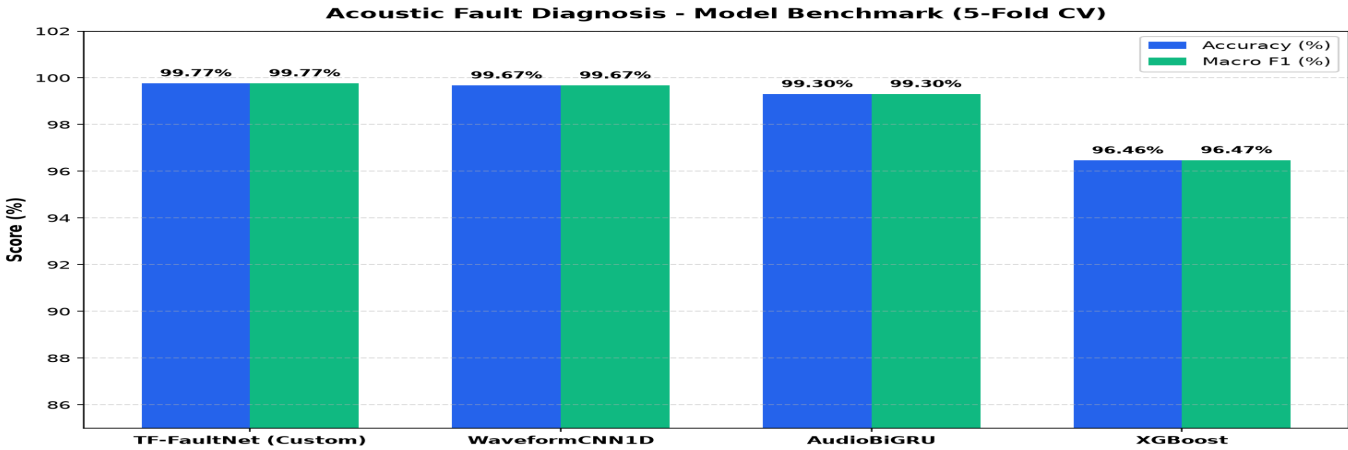


3. Detailed Per-Class Performance Breakdown (`TF-FaultNet`)

Fault Class / Mode	Physical Mechanism	Precision	Recall	F1-Score	Support
Normal (Normal Operation)	Clean Baseline Operation	100.00%	100.00%	100.00%	179
Escorregamento (Base Slippage)	Continuous Belt Friction	100.00%	99.44%	99.72%	179
Escorregamento_P1	Slippage on Pulley 1	98.89%	99.44%	99.16%	179
Escorregamento_P1P4	Slippage across Pulleys 1 & 4	100.00%	100.00%	100.00%	179
Perda_concentrada (Tooth Loss)	Impulsive Shock Impacts	100.00%	100.00%	100.00%	179
Perda_concentrada_P1	Tooth Loss on Pulley 1	99.44%	100.00%	99.72%	179
Perda_concentrada_P1P4	Tooth Loss across P1 & P4	100.00%	100.00%	100.00%	179
Perda_material (Wear / Erosion)	Diffuse Abrasive Wear	100.00%	99.44%	99.72%	179
Perda_material_P1	Material Wear on Pulley 1	100.00%	99.44%	99.72%	179
Perda_material_P1P4	Material Wear across P1 & P4	100.00%	99.44%	99.72%	179
Sem_P1 (Missing Pulley 1)	Missing Structural Pulley 1	100.00%	100.00%	100.00%	179
Sem_P1P4 (Missing P1 & P4)	Missing Structural Pulleys 1 & 4	98.90%	100.00%	99.44%	179

4. Multi-Criteria Architecture Selection Guide

- **Deploy `TF-FaultNet` (Recommended):** When running on Edge GPU/NPU (Jetson/Edge TPU) where 2D time-frequency attention and frequency explainability are paramount.
- **Deploy `WaveformCNN1D`:** When running on raw 1D audio streaming DSPs where FFT transformations cannot be computed.
- **Deploy `XGBoost`:** When deploying on low-power microcontrollers (PLC / ARM Cortex-M) with strict sub-millisecond CPU constraints (96.46% accuracy @ 0.45 ms).



Artifact References: MODEL_BENCHMARK_AND_ABLATION_COMPARISON.md | TF_FAULTNET_MASTER DISSERTATION.pdf | results/ablations/ablation_study_results.csv